New 42 Ma cratonic North American paleomagnetic pole from the Yukon underscores another Cordilleran paleomagnetism–geology conundrum

David T.A. Symons, Philippe Erdmer, and Phil J.A. McCausland

Abstract: Eocene posttectonic plutons of the Beaver River alkalic complex in southeastern Yukon intruded Devonian–Mississippian and Triassic sandstones in the Foothills of the Canadian Cordillera. A paleomagnetic collection of 27 sites from three separate plutons produced 326 specimens that were analyzed using alternating field and thermal step demagnetization methods. The A component characteristic remanent magnetization (ChRM) resides in magnetite with normal polarity in the 42.6 ± 0.8 Ma Beaver River pluton, reversed polarity in the 42.1 ± 0.7 Ma Larson Creek East pluton, and both polarities in the 41.3 ± 0.4 Ma Larson Creek West pluton, corresponding with magnetic polarity chrons 20n, 19r, and the boundary between chron 19r and 18n, respectively. The ChRMs of the plutons are indistinguishable (2σ) with a mean for the 42.0 ± 0.5 Ma complex of D = 158.8°, I = –73.1° (N = 21 sites, α_{95} = 3.0°, k = 116.8). A positive paleomagnetic contact test shows the A component to be primary, and the poorly isolated B component suggests the host rocks for Larson Creek West are Early to Middle Devonian. The paleopole for the Beaver River complex at 79.2°N, 145.8°E (N = 21, dp = 4.8°, dm = 5.4°; Q = 7) is concordant with interpolated 42 Ma reference poles for the North American craton. In contrast, paleopoles from the accreted Intermontane and eastern Coast Belt terranes record clockwise rotations of 24° ± 10° (Eocene) and 13° ± 5° (Oligocene–Pliocene), indicating that the allochthonous Intermontane terranes have been progressively driven ~240 ± 120 km eastwards up and over pericratonic and cratonic North American lower crust by Pacific plate subduction since the mid-Eocene.

Résumé : Des plutons post-tectoniques (Éocène) du complexe alcalin de Beaver River, dans le sud-est du Yukon, se sont introduits dans les grès (Dévonien–Mississippien et Trias) du piémont de la Cordillère canadienne. Trois cent vingt-six échantillons provenant d'une collection paléomagnétique de 27 sites de trois plutons distincts ont été analysés par les méthodes de champ alternatif et de démagnétisation thermique par étapes. La composante de la caractéristique A de l'aimantation rémanente (ChRM) réside dans la magnétite à polarisation normale dans le pluton de Beaver River, 42,6 ± 0,8 Ma, à polarité inversée dans le pluton de Larson Creek East, 42,1 ± 0,7 Ma et aux deux polarités dans le pluton de Larson Creek West, 41,3 ± 0,4 Ma, ce qui correspond aux chrones polaires magnétiques 20n, 19r et à la limite entre les chrones 19r et 18n respectivement. Les ChRM des plutons ne peuvent être distingués entre eux (2σ) avec une moyenne pour le complexe de 42,0 ± 0,5 Ma de D = 158,8°, I = –73,1° (N = 21 sites, α_{95} = 3,0, k = 116,8). Un test positif de contact paléomagnétique montre que la composante A est primaire et que la composante B, mal isolée, suggère que les roches encaissantes pour le pluton de Larson Creek West datent du Dévonien précoce à moyen. Le paléopôle pour le complexe de Beaver River à 79,2° N, 145,8° E (N = 21, dp = 4,8°, dm = 5,4°; Q = 7) concorde avec les pôles de référence interpolés de 42 Ma pour le craton de l'Amérique du Nord. Par contre, les paléopôles des terranes accrétés du Plateau intérieur et de la Ceinture côtière Est enregistrent des rotations dans le sens des aiguilles d'une montre de 24° ± 10° (Éocène) et de 13° ± 5° (Oligocène–Pliocène), indiquant que les terranes allochtones du Plateau intérieur ont été progressivement amenés ~240 ± 120 km vers l'est au-dessus de la croûte inférieure, péricratonique et cratonique, du Bouclier canadien par la subduction de la plaque du Pacifique depuis l'Éocène moyen.

[Traduit par la Rédaction]

Introduction

Reliable paleomagnetic estimates of accreted terrane tectonic motion require reliable cratonic reference poles for comparison. To date, such estimates have been made for the accreted terranes of the North American Cordillera using paleomagnetic poles from some two hundred Mesozoic and Cenozoic rock units. The tectonic estimates have been done by comparing a rock

Received 5 December 2002. Accepted 3 June 2003. Published on the NRC Research Press Web site at http://cjes.nrc.ca on 15 October 2003.

Paper handled by Associate Editor F. Cook.

D.T. Symons[1] and P.J.A. McCausland.[2] Department of Earth Sciences, University of Windsor, Windsor, ON N9B 3P4, Canada.
P. Erdmer. Department of Earth and Atmospheric Sciences, University of Alberta, Edmonton, AB T6G 2E3, Canada.

[1]Corresponding author (e-mail: dsymons@uwindsor.ca).
[2]Present address: Department of Geological Sciences, University of Michigan, Ann Arbor, MI 48104, U.S.A.

 doi: 10.1139/E03-047 © 2003 NRC Canada

unit's pole to (1) a selection of cratonic poles of similar age or (2) an interpolated pole from a published apparent polar wander path (APWP) for the North American craton. The former method has the disadvantage that the translation and rotation estimates depend in part on the selected cratonic poles and their weighting. For example, Johnston et al. (1996) selected five poles to represent the Late Cretaceous North American reference pole to compare with the 70 Ma Carmacks Group volcanics' pole. Their estimated northward translation for the Intermontane Belt terranes of ~1900 ± 700 km would increase to ~2400 ± 700 km had the reference pole been derived from the APWPs of Besse and Courtillot (1991) or Van der Voo (1993). The latter method is less subjective and more amenable to comparison with other tectonic motion estimates (e.g., Harris et al. 1999), but the investigator must choose which current APWP to use, recognizing that each path is based on the paleomagnetic and age data available at the time, on the data selection, and on the subsequent methods used to obtain the path.

This paper reports paleomagnetic data from three mid-Eocene (~42 Ma; Bartonian) alkalic syenite intrusions from the Beaver River complex that were emplaced into the North American craton in easternmost Yukon after Laramide deformation (Fig. 1). The Beaver River pole is compared to current APWPs for the North American craton that rely on just six poles from three restricted localities for the 37–47 Ma interval, i.e., the Mistastin Lake volcanics in Labrador (Currie and Larochelle 1969), two poles from the Monterey intrusions in the Virginias (Lovlie and Opdyke 1974; Ressetar and Martin 1980) and three volcanic sequences in Wyoming (Shive and Pruss 1977; Sheriff and Shive 1980; Flynn 1986). Our result also has implications for the timing and extent of final motions of the adjacent accreted Intermontane Belt terranes relative to North America.

Geology

Three syenitic intrusions of < 1km in diameter in the Foreland Belt of the Cordillera in the southwest Yukon (National Topographic System (NTS) map areas 95C/3,C/6) form part of the Tertiary to Recent Beaver River Alkaline Complex. Known as the Larsen Creek East, Larsen Creek West, and Beaver River syenite bodies (Rowins and Francis 1995; Fig. 2), their mineralogy is dominated by amphibole and orthoclase with minor clinopyroxene, biotite, nepheline, sodic plagioclase, magnetite, and apatite. The Larsen Creek East and West plutons are compositionally zoned with mafic centres and felsic margins. The rocks are strongly Si-undersaturated, sodic A-type syenitoids, whose bulk composition and trace-element characteristics indicate a single high-level magma chamber. The zoning implies emplacement of progressively more mafic batches of magma into the pluton's centre. The Beaver River syenite is unzoned but lacks data from its marginal rock type(s). All members of the complex are interpreted to be from similar upper mantle sources (Rowins and Francis 1995).

Unpublished Ar^{40}/Ar^{39} whole-rock plateau dates indicate cooling of Larsen Creek East at 42.1 ± 0.7 (1σ) Ma, Larsen Creek West at 41.3 ± 0.4 Ma, and Beaver River at 42.6 ± 0.8 Ma (D.M. Francis, personal communication, 1999). Nearby is an Eocene (?) biotite syenite pluton, an intruding aphanitic breccia (Pigage and Allen 2001), and a clinopryroxene melanephelinite spatter cone (Rowins and Francis 1995). Nearby also is the Precambrian–Cambrian Pool Creek syenite, providing a curious record of episodic alkaline magmatism spanning > 500 million years within a 500 km^2 area.

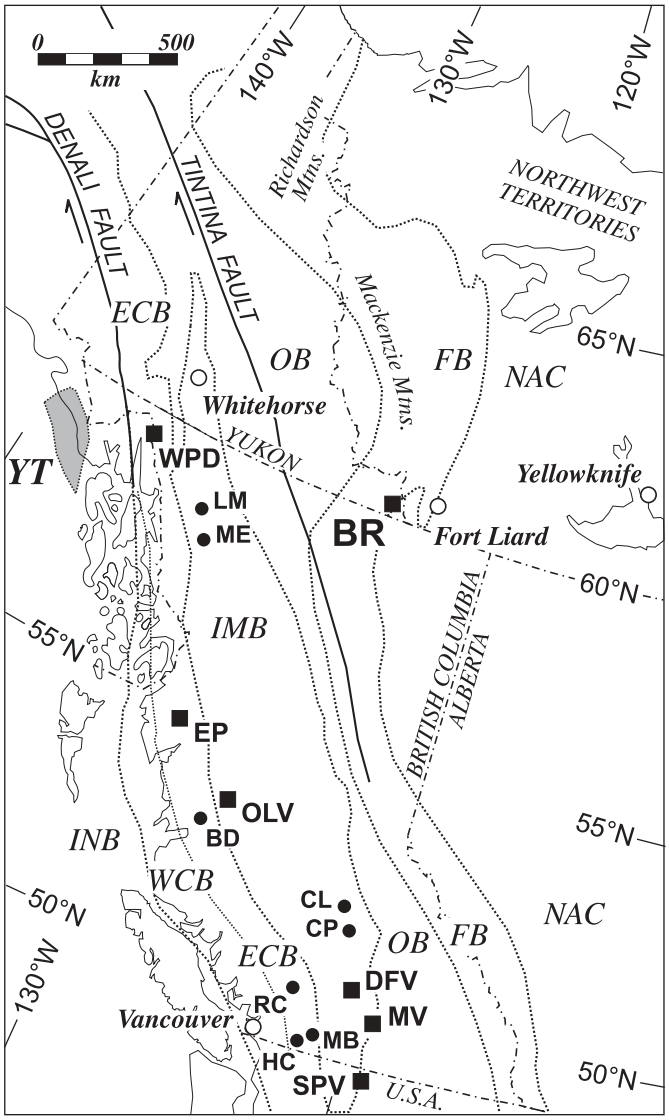
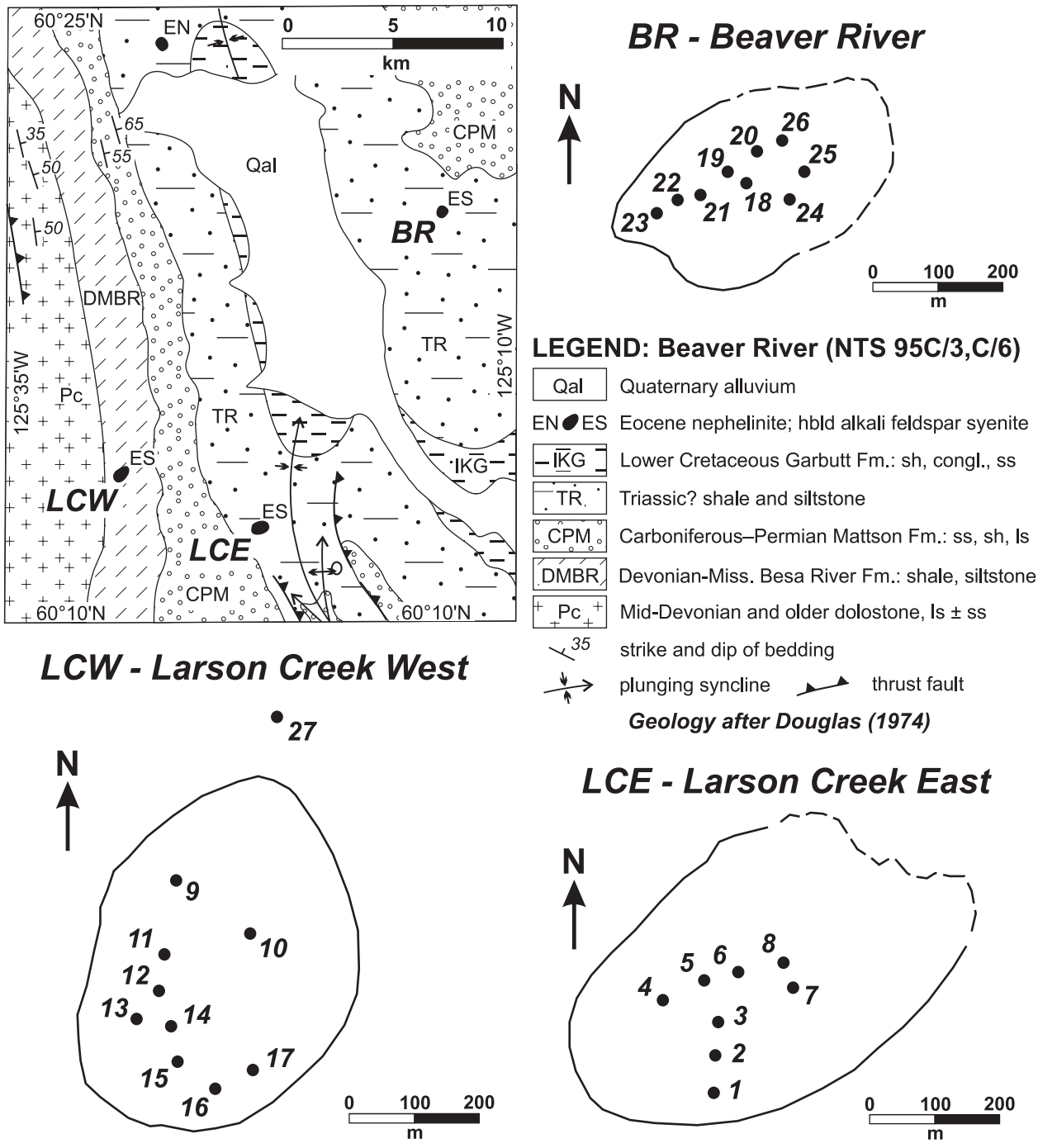


Fig. 1. Cordilleran Orogen of Canada. Morphogeological belts: ECB, Eastern Coast Belt; FB, Foreland Belt; IMB, Intermontane Belt; INB, Insular Belt; NAC, North American craton; OB, Omineca Belt; WCB, Western Coast Belt. YT is the Yakutat terrane. BR, Beaver River study area. Other Eocene paleomagnetic studies: DFV, Dewdrop Flats volcanics; EP, Eocene plutons; MV, Marron volcanics; OLV, Ootsa Lake volcanics; SPV, Sanpoil volcanics; WPD, White Pass dikes (see text for references). Paleomagnetic studies on ≤42 Ma units (Table 3): BD, Brown mafic dikes; CB, Chiliwack batholith; CL, Cariboo lavas; CP, Cariboo mafic plugs; HC, Hope complex; LM, Level Mountain lavas; MB, Mt. Barr pluton; ME, Mt. Edziza lavas; RC, Rogers Creek pluton.

Host rocks of the plutons are Devonian–Mississippian and Triassic fine-grained clastic sedimentary strata (Douglas 1976). Folded into upright open folds at the regional scale and cut by widely spaced, small-displacement thrusts, their structural style is that of the northern Foothills subprovince of the

© 2003 NRC Canada

Fig. 2. Geology of the study area with sketch maps of the sampled plutons. Fm., Formation; Miss., Mississippian; NTS, National Topographic System.



Foreland Belt in adjacent British Columbia, where the dominant folds formed above a regional detachment in the Upper Devonian and Mississippian shales of the Besa River Formation (McMechan and Thompson 1989). Folds and faults along the eastern edge of the northern Foothills deform the youngest Late Cretaceous strata: consistent, by analogy with areas along strike to the north and south, with Early to Middle Eocene regional compressive deformation. Noting their cooling ages, the sampled intrusions are inferred to postdate regional (Laramide) deformation entirely. Because of extensive drift cover, bedding attitudes in the host rocks are difficult to determine. Regionally, the bedding dips moderately eastward over 40 km in three west-vergent thrust slices (Douglas 1976). Axial projection from the Pool Creek map area to the immediate northwest (Pigage and Allen 2001) also suggests that the bedding dips ≤30° eastwards. Approximately 15 km southeast along strike of the Larsen Creek East intrusion, a hanging-wall anticline affects Carboniferous strata above a west-vergent thrust and the bedding locally dips westward.

© 2003 NRC Canada

Experimental methods and results

Sampling

The plutons were accessed by helicopter from Fort Liard, N.W.T. (Fig. 1). Twenty-seven sites were collected: eight from the Larson Creek East pluton, nine from the Larson Creek West pluton, one from the host rock close to its contact, and nine from the Beaver River pluton (Fig. 2; Table 1). In each pluton, the sites were distributed uniformly throughout the available outcrops. Six or seven cores were drilled and then oriented in situ at each site. Two, but sometimes one or three, specimens were prepared from each core. A total of 326 specimens were analysed paleomagnetically after storage for several months in a magnetically shielded room to allow their viscous remanent magnetization (VRM) to substantially decay.

Step demagnetization

Initially, the natural remanent magnetizations (NRM) of the specimens were measured on an automated Canadian Thin Films CTF-420 cryogenic magnetometer. The median NRM intensities for the Larson Creek East, Larson Creek West, and Beaver River plutons and host rock specimens were 9.7×10^{-1} A/m, 1.8×10^{-1} A/m, 2.7×10^{0} A/m, and 5.1×10^{-4} A/m, respectively. One pilot specimen from each site was alternating field (AF) demagnetized in 10 steps up to 140 mT using a Sapphire Instruments SI-4 AF demagnetizer. Larson Creek East specimens showed the initial removal of a downwardly directed low-coercivity VRM component up to ~20 mT and then the isolation of a steeply upward or reversed-polarity magnetization to the south-southeast until 140 mT (Fig. 3*a*). This characteristic remanent magnetization (ChRM) is termed the Ar component. In contrast, the Beaver River specimens exhibited very little VRM that was removed by 5 or 10 mT. Thereafter, they displayed a downward or normal polarity ChRM, antiparallel to Ar, that is termed the An component (Fig. 3*b*). Larson Creek West specimens proved to be far less consistent with substantial VRM components up to 50 mT or higher. Their ChRMs included An, Ar, and other components.

A second pilot specimen from each site was demagnetized in 12 steps up to 600 °C using a Magnetic Measurements MMTD-1 oven. Larson Creek East specimens showed removal of a VRM component up to about 300 °C and the isolation thereafter up to 580 °C of the Ar component (Fig. 3*c*). The linear nature of the ChRM intensity curve on demagnetization from 400 °C to 580 °C indicates that magnetite is the sole significant remanence carrier. Beaver River specimens behaved similarly on thermal demagnetization but gave the An direction (Fig. 3*d*). Again, the Larson Creek West specimens yielded variable results with their VRM retained at higher temperatures and their ChRM entirely removed by lower maximum temperatures. Ar, An, and other components were isolated by thermal step demagnetization, and magnetite or titanomagnetite was the sole significant remanence carrier. Based on results from data collected from the pilots, the remaining plutonic specimens were AF demagnetized in six or seven steps between 15 to 20 mT and 80 to 120 mT to isolate their ChRM. The results from the host rock at site 27 will be discussed in section 3.5.

Data analysis

The specimen ChRM directions were determined using orthogonal vector component plots (As and Zijderveld 1958), the principal component method (Kirschvink 1980), and the remagnetization circle method (Bailey and Halls 1984). Site and unit mean ChRM directions were calculated following Fisher (1953).

For the Larson Creek East and Beaver River plutons, most specimens — 72% and 95%, respectively — gave well-defined endpoint ChRM directions (Table 1). Excluding site 25, they gave well-defined Ar and An pluton mean directions, respectively. At site 25, the specimens include one Ar endpoint, two An endpoints, and the other specimen directions are strung out roughly along the intervening great circle, hence the site mean ChRM direction is not included in the pluton's direction or overall mean ChRM direction. Although 52% of Larson Creek West specimens give defined endpoints, the majority come from sites 9, 13, and 17, with mean directions falling roughly on a great circle between the Ar and An directions (Fig. 4*a*). Also all sites from this pluton retained both Ar and An specimen directions (Table 1). This implies that the emplacement and cooling of the Larson Creek West pluton spanned a polarity transition of the Earth's magnetic field, probably from normal to reversed polarity because Ar sites 10, 11, 12, and 14 are closest to the pluton's core, whereas An site 16 is the most peripheral site, and transitional sites 9, 13, and 17 are in between.

When the site mean directions are grouped, the plutons have similar mean directions, excepting polarity, that are not statistically distinct from each other at 95 % confidence (Table 1; Fig. 4*b*). Combining all site mean directions, after converting the normal sites to their antipodes, gives a mean for this Eocene intrusive event of $D = 158.8°$, $I = -73.1°$ ($N = 21$, $\alpha_{95} = 3.0°$, $k = 116.8$) (Table 1).

Saturation isothermal remanence

Saturation isothermal remanent magnetization (SIRM) analyses were run on three specimens from each pluton. The samples were pulse magnetized in 15 direct field steps up to 900 mT using a Sapphire Instruments SI-6 pulse magnetizer and then AF demagnetized in 10 steps up to 160 mT. All specimens reached 90 % of saturation between 60 and 150 mT, and then climbed slowly to 100 % at 900 mT, suggesting that their ChRM is carried by a mixture of single, pseudosingle, and multidomain magnetite. This interpretation is consistent with the results from AF demagnetization of the specimens' SIRM, which show a rapid intensity decrease by 50 to 120 mT and then a slow decrease of the remaining few percentage of the SIRM intensity, and with the thermal step demagnetization tests described previously.

Contact test

The clastic host rocks around the plutons outcrop sparsely. However, one outcrop of silicified sandstone was found amid the scree at Larson Creek West (site 27, Fig. 2) with a bedding strike of N55°E and dip of 12°SW. It was sampled for a paleomagnetic contact test (Everitt and Clegg 1962). The test asserts that the pluton carries primary ChRM if (*a*) its direction is the same as that of its baked contact zone but differs from that of more remote host rock, or (*b*) its direction is found in the lower coercivity and unblocking temperature

© 2003 NRC Canada

Table 1. Mean Remanence Directions.

Site, Pluton	Specimens (sites)				Characteristic Remanence				
	*N*m	*N*e	*N*r	*N*u (*N*s)	*D* (°)	*I* (°)	α_{95}(°)	*k*	Polarity
01 LCE	12	11	0	11	175.9	–57.6	5.1	80.3	–
02 LCE	13	13	0	13	175.7	–74.9	3.1	177.6	–
03 LCE	11	10	0	10	159	–70.9	9.6	26.3	–
04 LCE	10	7	3	10	135	–74.2	6.4	58.4	–
05 LCE	11	7	4	11	153.7	–75	5.5	70	–
06 LCE	12	11	1	12	179.3	–69.3	3.3	172.2	–
07 LCE	12	7	4	11	172.2	–70.4	5.7	64.1	–
08 LCE	11	4	6	10	175.6	–72.1	6.7	52.3	–
09 LCW	12	12	0	12	313.8	–62.3	13.2	11.8	M, 2
10 LCW	13	6	5	11	166.5	–71.7	5.1	81.3	M, 6
11 LCW	12	4	8	12	169.3	–78.1	5.6	61.1	M, 3
12 LCW	11	1	10	11	165.4	–78.3	5.8	63.8	M, 1
13 LCW	11	11	0	11	244	–30.7	20	6.2	M, 2
14 LCW	12	3	8	11	170.7	–78	3.1	212.8	M, 2
15 LCW	14	0	0	0					
16 LCW	12	9	0	9	324.2	71.5	7.5	48.6	M, 1
17 LCW	14	12	0	12	286.9	–5.4	9	24.3	M, 0
18 BR	12	12	0	12	351.9	76.3	1.9	523.4	+
19 BR	12	12	0	12	349.7	77.4	1.6	736	+
20 BR	15	15	0	15	344.2	75.6	3.9	97	+
21 BR	12	12	0	12	342.4	66.2	3.5	156.6	+
22 BR	14	14	0	14	300.1	64.9	3.8	107.9	+
23 BR	13	11	2	13	334	74.8	2.7	236.3	+
24 BR	11	11	0	11	315.7	68.5	5	84.6	+
25 BR	9	9	0	9	286.6	7.3	28.8	4.2	M, 1
26 BR	12	9	1	10	306.4	71.5	9.9	24.8	+
27 H lo	10	9	0	9	331.4	77.1	8.7	36.1	+
27 H hi	10	9	0	9	154.3	16.1	21.7	6.6	M, 3
LCE	(1 – 8)			8	167.5	–71.1	5.1	120.2	–
LCW*	(10 – 12 14 16)			5	161.9	–75.7	4.4	309.7	M
BR	(18 – 24 26)			8	327.5	72.9	5.5	103.9	+
Unit*	(LCE + LCW + BR)			21	158.8	–73.1	3	116.8	

Note: Plutons: LCE, Larson Creek East; LCW, Larson Creek West; BR, Beaver River. Specimens: *N*m, measured; *N*e endpoints; *N*r, remagnetization circles; *N*u, used for statistics. Characteristic Remanence: *D*, declination (in degrees); *I*, inclination (in degrees); α_{95}, radius of cone of 95 % confidence (in degrees); *k*, precision parameter (Fisher 1953). Polarity: +, positive; –, negative; M (followed by number of antipodes included in mean), mixed; *N*s, number of site mean directions averaged; * antipodes of site mean directions used.

domains of the host rock but differs from that found in higher domains. The latter case is seen at site 27.

The site 27 specimens were AF step demagnetized up to 130 mT, reducing the NRM intensity of 2 to 20 × 10^{-4} A/m by 50% ± 25% and defining an An direction similar to that found at site 16 of the pluton's perimeter (Fig. 5). Their remaining remanence was thermally demagnetized at 500, 530, and 560 °C, isolating a poor clustering of mixed-polarity, shallow-inclination ChRM directions in the 1–6 × 10^{-5} A/m range that are directed towards the south-southeast or north-northwest, and labelled the Bm component (Fig. 5).

SIRM analysis of host rock specimens produced a rapid intensity acquisition up to ~50% of the SIRM value by 275 mT and then a steady linear climb to 100 % at 900 mT. AF demagnetization of the SIRM led to ~50% intensity drop by 160 mT, indicating that the NRM is carried about equally by magnetite and hematite. These results, plus the AF and thermal demagnetization results, indicate that the sandstone's An ChRM component is secondary and carried by magnetite, whereas its Bm component is either primary or early diagenetic and carried by hematite.

After tilt correction, the An direction of the host rocks becomes almost vertical, so that its virtual pole is only 3° from the sampling site. This impossible answer further indicates that the intrusion postdates all Laramide deformation. The Bm component defines, albeit imprecisely, a Lower to Middle Devonian pole position both before and after tilt correction on the APWP of Van der Voo (1993) and is deemed to be a residual primary remanence component.

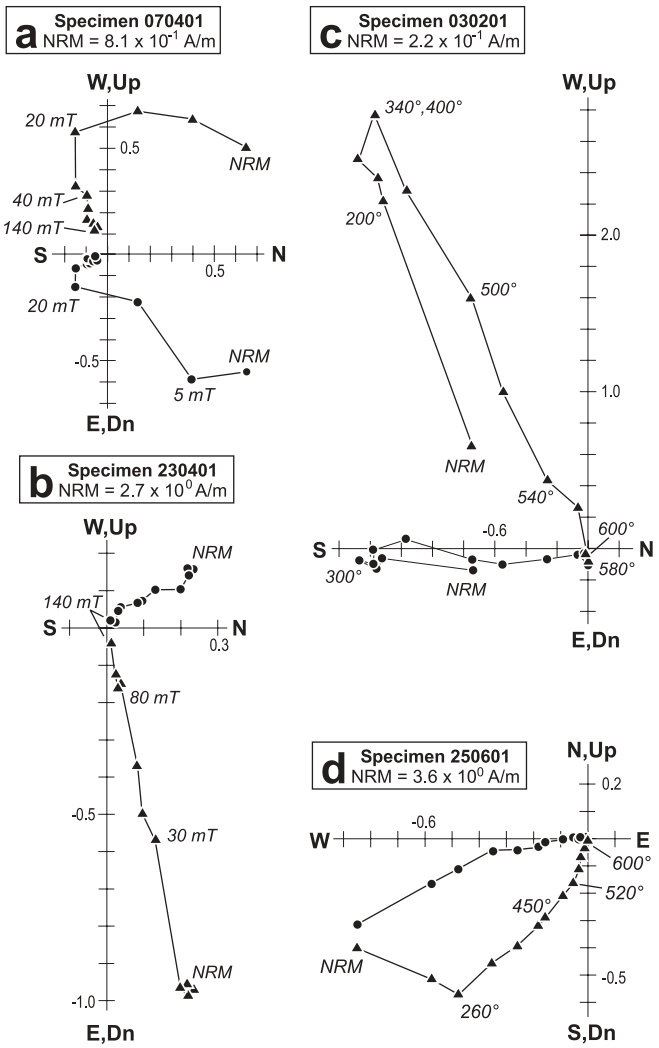
Discussion

Beaver River pole

Using the conventional geocentric axial dipole (GAD) model for the Earth's magnetic field, the Beaver River plugs yield a pole position of 79.2°N, 145.8°E (*N* = 21, dp = 4.8°, dm = 5.4°)

© 2003 NRC Canada

Fig. 3. Vector component plots for example specimens showing (*a*) the Ar component from site 7, and (*b*) the An component from site 23, isolated by alternating field (AF) step demagnetization; and (*c*) the Ar component from site 3, and (*d*) the An component from site 25, isolated by thermal step demagnetization. Circle symbols show the remanence component in the horizontal plane, triangle symbols show it in vertical plane. Plots (*a*)–(*c*) use projections onto the north–south (N–S) vertical plane, whereas plot (*d*) is projected onto the east–west (E–W) vertical plane. Axial intensities are normalized to the NRM intensity, with its value listed for each specimen. AF steps are in mT and thermal steps in °C.

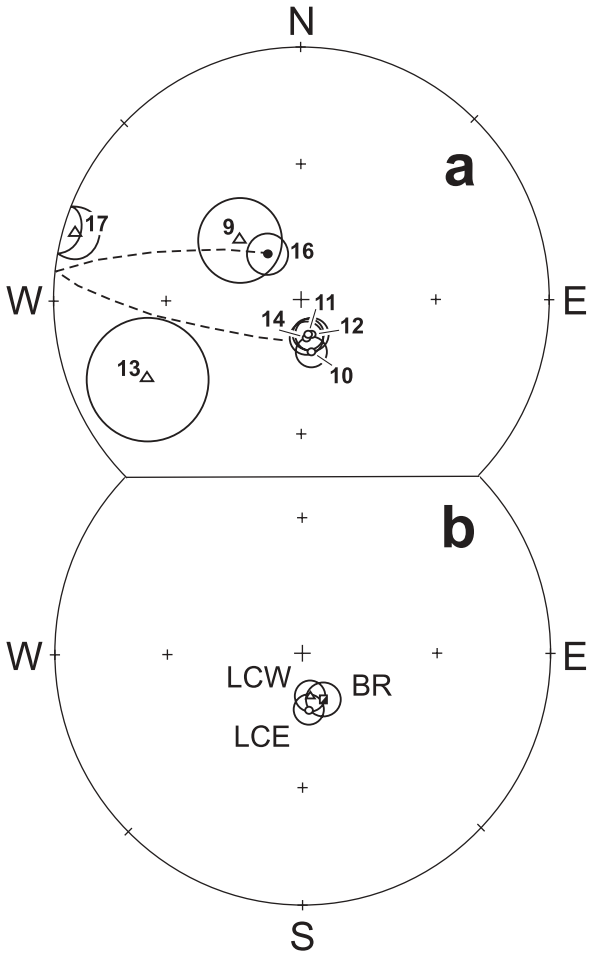


(latitude, longitude (number of sites represented, radii of the oval of 95% confidence along and perpendicular to the site-pole great circle)). The age of the pole is the average and root mean square deviation obtained from the three Ar^{40}/Ar^{39} age dates 42.0 ± 0.5 Ma.

Reversal chronology

The ChRM data for the three plutons provide an excellent fit to the magnetic polarity time scale of Cande and Kent (1995). On it, the normal-polarity Beaver River falls in normal chron 20n, the reverse-polarity Larson Creek East falls in reverse chron 19r, and the normal-to-reverse transitional-polarity Larson Creek West falls on the boundary between chrons 19n and 18r (Fig. 6). The high quality of the fit validates the accuracy of both the Ar^{40}/Ar^{39} age dates and the magnetic polarity time scale, and it affirms that a primary ChRM has been recorded with fidelity in these plutons.

Fig. 4. Equal-area stereonet showing (*a*) site mean directions for the Larson Creek West pluton, with An (●), Ar (○), and transitional (△) components, and (*b*) pluton mean directions for the Beaver River (BR), Larson Creek East (LCE) and Larson Creek West (LCW) plutons with their cones of 95% confidence, using antipodes for normal-polarity sites. Solid (open) symbols indicate normal or downward (reversed or upward) remanence directions.



North American reference pole at 42 Ma

Besse and Courtillot (1991) selected 121 poles from geologic units for the 0 to 200 Ma time interval from the Eurasian (23), African (35), North American (51) and Indian (2) plates that they deemed to be most reliable. Using a 20 Ma sliding window, they generated an APWP for each plate, combined the paths into a master path on a common reference frame using plate kinematic models derived from sea-floor magnetic anomalies, generated reference poles at 10 million year intervals on the master path, and then reversed the model rotations to obtain a synthetic APWP for each plate. For the 42 ± 10 Ma interval, their path utilizes six North American poles and one European pole, but none are in the 42 ± 5 Ma

© 2003 NRC Canada

Fig. 5. Equal-area stereonet for contact test site 27, showing (*a*) AF demagnetization for two example specimens from 0 to 130 mT to isolate the An component and then thermal step demagnetization from 500 °C to 560 °C to isolate the Bm component specimen directions; and (*b*) all An and Bm specimen directions with their mean (larger symbol) and cone of 95% confidence. Symbol conventions as in Figs. 3 and 4.

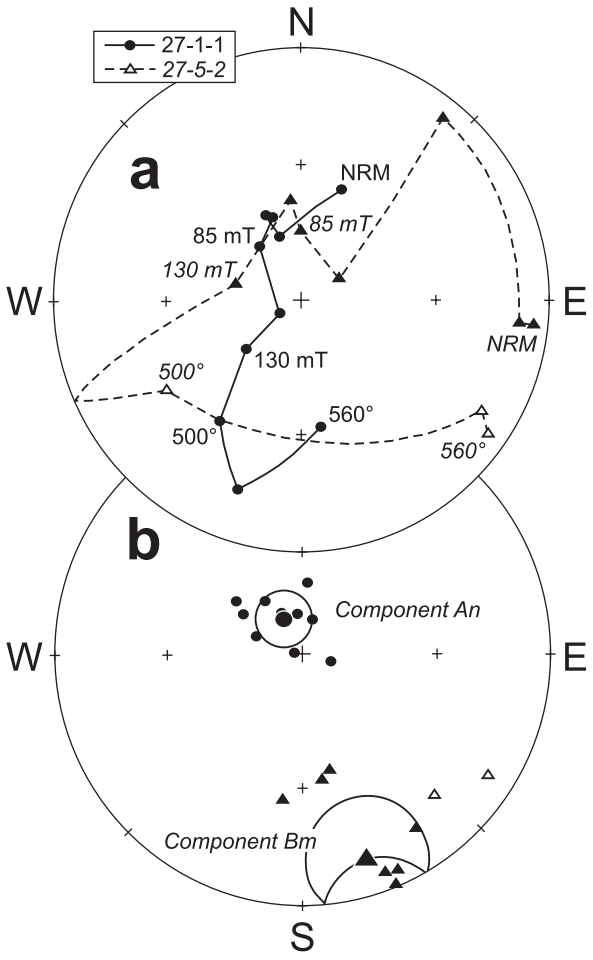
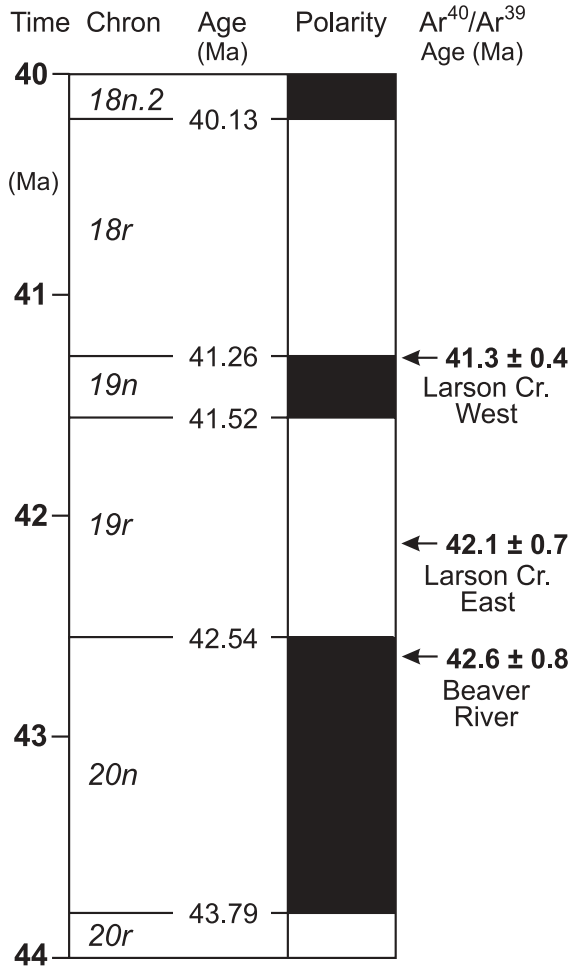


Fig. 6. Ar^{40}/Ar^{39} whole-rock plateau ages for the three Beaver River alkalic complex plutons on the magnetostratigraphic time scale (Cande and Kent 1995). Solid or "n" (unfilled or "r") intervals are normal (reversed) in magnetic polarity. Cr., Creek.



interval. Their first reference pole (Model BC1, Table 2) interpolates between their 40 Ma and 50 Ma reference poles, and the second (Model BC2, Table 2) interpolates between 42.8 and 27.9 Ma mean ages for the poles that bracket 42.0 Ma.

Van der Voo (1993) introduced a system for assessing the quality (Q) of a pole from a rock unit based on seven reliability criteria, giving a scale for Q from 0 to 7. Accepting Phanerozoic poles with $Q \geq 3$ from North America, he grouped them into intervals of ~ 25 Ma duration (table 5.1 in Van der Voo 1993). His youngest interval includes 17 Paleocene and Eocene poles for North America (Model V1, Table 2) or 26 poles if nine Greenland poles are rotated into North American co-ordinates (Model V2). In general, these poles have a mean age of ~54 Ma and thus match the 42 Ma Beaver River pole poorly. Also, this path has been superceded by the following Torsvik et al. (2001*b*) analysis.

McElhinny and McFadden (2000, their table 6.5) used the $Q \geq 3$ criteria of Van der Voo (1993) on an updated listing of Early Cambrian to Eocene poles, grouping them by geological periods of ~25 Ma. Deeming each site mean direction to measure the Earth's GAD when averaged over 25 Ma, they generated an artificial Fisher (1953) population of site mean directions for each rock unit's pole and combined them to obtain the reference poles. Thus their North American Eocene reference pole incorporates 401 site mean directions from 14 pole populations (Model MM1, Table 2). Assuming a linear rate of polar wander from it at 45 Ma to the present North Pole at time zero, the interpolated 42 Ma reference is Model MM2 (Table 2).

Torsvik et al. (2001*a*, 2001*b*) illustrate well how different calculation methods lead to different reference poles despite using the same data set of poles. Using poles with $Q \geq 3$ and ages between 40 and 130 Ma from North America, Greenland, Svalbard,and Europe, they used four methods to calculate the reference poles from which we have interpolated between the 45 Ma and 40 Ma poles. The fact that the data set is truncated at 40 Ma is a concern because 30 to 39 Ma poles are not included in their window average.

Model Trd (Torsvik et al. 2001*b*) assumes the Earth's magnetic field can be represented by a GAD, that European poles can be rotated into North American co-ordinates by closing the Atlantic Ocean based on sea-floor magnetic anomaly data and the hotspot reference framework, and that poles

© 2003 NRC Canada

Table 2. Estimates from APWPs of the North American reference pole at 42 Ma.

	Reference Pole				
Model	°N	°E	$°A_{95}$	*N*p	Reference
BC1	78.8	160	3.9	7	Besse and Courtillot (1991): interpolated within 40–50 Ma
BC2	79.4	159	3.7	7	Besse and Courtillot (1991): interpolated between 42.8 and 51.7 Ma
V1	82	168	5	17	Van der Voo (1993): Eocene, North American poles only
V2	77	173	4	26	Van der Voo (1993): including rotated Greenland poles
MM1	80	158	3.7	14	McElhinny and McFadden (2000): 45 Ma mean
MM2	80.7	158	3.7	14	McElhinny and McFadden (2000): interpolated to 42 Ma
Trd	81.5	165.4	4.5	9, 18*	Torsvik et al. (2001*a*): interpolated in 40–45 Ma window
Tspl	84.7	129.8	—	9, 18*	Torsvik et al. (2001*a*): interpolated from spline fit APWP
Tna	83.4	152.5	4.5	7, 11*	Torsvik et al. (2001*a*): using North American poles only
T′	82.8	201.7	4.1	9, 18*	Torsvik et al. (2001*b*): from 8% octupolar APWP
BR	79.2	145.8	5.1	—	This study: cratonic pole from this study
BR′	79.4	176.4	5.4	—	This study: calculated including an 8% octupolar contribution

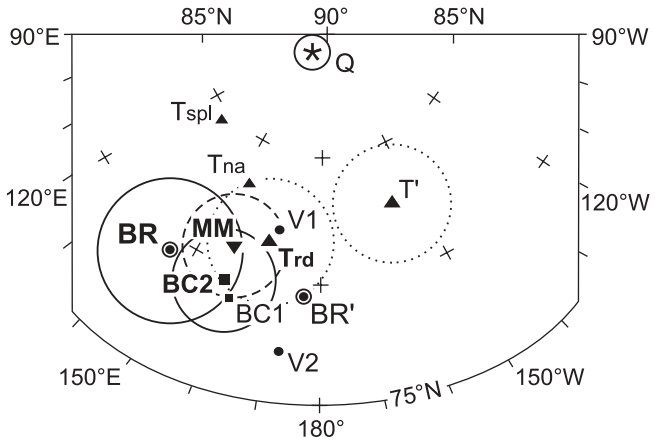
Note: Model: see text for the basic assumptions built into the path construction and how 42Ma was determined from it. **Boldface** models are preferred, as discussed in the text and as shown in Fig. 7. Reference Pole: °N, latitude (in degrees); °E, longitude (in degrees); $°A_{95}$, radius of cone of 95% confidence (in degrees); *N*p, number of unit poles used to calculate the mean, where * denotes the number in the 40 Ma and 45 Ma bracketing poles, respectively.

within a 20 Ma moving window can be averaged to obtain the reference poles (Table 2). Model Tspl (Torsvik et al. 2001*b*) differs from Model Trd in that the reference poles were obtained by fitting a spherical spline to the data set with the poles weighted according to their *Q* and α_{95} values. Thus the most reliable poles contribute most to the reference poles but the amount of weighting is subjective. Model Tna (Torsvik et al. 2001*b*) differs from Model Trd in that only North American poles are used. This eliminates errors in the parameters for rotating European poles into North American co-ordinates but decreases the available number of unit poles so that the pole's cone of 95% confidence will usually increase. Model T′ (Torsvik et al. 2001*a*) is the same as Model Trd, except that the fit to the data set has been optimized to include an 8% octupole contribution to the GAD to describe the Earth's field.

Note that, with two exceptions, all of the Eocene poles utilized by Torsvik et al. (2001*a*, 2001*b*) were published when Besse and Courtillot (1991) generated their APWP, i.e., all 42 Ma reference poles considered here came from virtually the same data set.

Shown in Fig. 7 are the 42 Ma reference poles and the Beaver River pole (Table 2). Some of the poles are circumscribed by their 1σ cone of confidence. To a close approximation, if the 1σ cones of two poles do not overlap, then the poles are statistically different at the 95% (2σ) confidence level. Evidently, the pole for the Beaver River alkalic complex is concordant with Model BC1 and our preferred Model BC2 based on the Besse and Courtillot (1991) APWP and with Model MM1 and our preferred Model MM2 based on the McElhinny and McFadden (2000) path. Although Models V1 and V2 from Van der Voo (1993) are superceded by Torsvik et al. (2000*b*) for Cenozoic time, it is apparent that the Model V1 pole is concordant with the Beaver River and the Model BC1, BC2, MM1, and MM2 poles, whereas the Model V2 pole is significantly different, suggesting problems with either the Greenland poles, several of which have very broad error limits on their age, or with the rotation parameters used to rotate

Fig. 7. Equal-area polar-projection showing the reference poles described in the text (Table 2), and determined from the APWPs of Besse and Courtillot (1991) (BC1 and BC2), Van der Voo (1993) (V1, V2), McElhinny and McFadden (2000) (MM covers both MM1 and MM2), Torsvik et al. (2001*b*) (Trd, Tna,Tspl) and Torsvik et al. (2001*a*) (T′). BR and BR′ are the Beaver River alkalic complex poles calculated using the geocentric axial dipole (GAD) and 8% octupole methods, respectively. Pole *Q* is from 0 to 5 Ma igneous rocks (McElhinny et al. 1996). The BR and some reference poles are circumscribed by their 1σ cone of confidence.



them into the North American framework (cf. Torsvik et al. 2001*b*).

The Beaver River pole best fits the preferred 42 Ma Model Trd pole of the Torsvik et al. (2001*a*) alternatives, although the agreement with the Model Tna pole is also statistically significant. A statistical comparison cannot be made with the Model Tspl pole because confidence limits cannot be calculated for it, however the correlation appears to be marginal, likely due to the previous lack of a reliable 42 Ma age unit pole to guide the spline fit. The Model T′ pole, which includes an 8% octupole contribution to the Earth's GAD magnetic field, appears to be a poor fit to the Beaver River pole. However, it

© 2003 NRC Canada

is no longer significantly different upon recalculating the Beaver River pole to include an 8% octupole contribution.

Our preferred Model BC2, MM2, and Trd poles are indistinguishable from each other at > 99% confidence, as shown by their mean of 160.6°E, 80.5° (N = 3; A_{95} = 1.9°; k = 4287), and from the Beaver River pole. This indicates that the Beaver River pole has averaged out bias from secular variation in the Earth's magnetic field and that the plutons were not tilted after intrusion. Inserting the Beaver River pole into the calculation for each of the preferred reference poles changes them non-significantly by less than 0.1° of arc.

Implications for tectonic evolution of the Canadian Cordillera

Geophysical evidence for eastward translation

The Beaver River paleopole provides a local cratonic reference point with no evident translation or rotation to compare with poles from the accreted terranes of the northern Cordillera to the immediate west. Recently Symons et al. (2000) summarized Eocene paleomagnetic results from the accreted terranes of the Intermontane Belt and the eastern Coast Belt (Fig. 1). In Canada, these belts extend for ~1900 km along the western margin of the North American craton from Washington State into Alaska. With an average width of ~300 km, they extend from the Tintina – Northern Rocky Mountain Trench on the eastern side, about 200 km west of the study area across the Foreland Belt, to the Denali – Coast Shear Zone on the western side. Symons et al. (2000) calculated tectonic translation and rotation estimates using the Besse and Courtillot (1991) cratonic APWP for the 51.5 ± 1.5 Ma Sanpoil volcanics (Fox and Beck 1985) and the 52 ± 2 Ma Marron volcanics (Bardoux and Irving 1989) in the Quesnel terrane, the 49.4 ± 2.4 Ma Dewdrop Flats volcanics (Symons and Wellings 1989) and 49.6 ± 1.7 Ma Ootsa Lake volcanics (Vandall and Palmer 1990) in the Stikine terrane, and the 46 ± 2 Ma Eocene plutons near Kitimat (Symons 1974; 1977) and 51 ± 1 Ma White Pass dikes near Skagway (Symons et al. 2000) that were emplaced into the margin of the Intermontane Belt. Two fundamental conclusions were drawn. First, as the 1989 studies above concluded, the Intermontane Belt and eastern Coast Belt terranes had halted northward translation relative to the craton by 51 Ma. Second, and more curious given the geographic spread of studies, they record continuing clockwise rotation by 24° ± 10° on average after 51 Ma, leading Symons et al. (2000) to speculate that rotation was occurring on underlying sole faults. Three recent lines of evidence provide further support for this hypothesis.

First, in addition to the Lithoprobe deep crustal profile across the southern Canadian Cordillera that appears to have imaged subhorizontal sole or décollement faults (Cook 1995), similar faults have now been imaged at mid crustal levels on two Lithoprobe profiles across the accreted Intermontane terranes in northern British Columbia and the Yukon (Snyder et al. 2002).

Secondly, from geothermal modeling based on numerous heat flow measurements in northern British Columbia and the Yukon, Lewis and Hyndman (2001) and Lewis et al. (2002) have determined that only the upper crust has sufficient lithospheric strength to transmit stress. They speculate that the obliquely converging Pacific plate is pushing cold, quasi-rigid Cordilleran continental crust over a hot weak lower crust on detachment faults across the accreted terranes to the Foreland Belt, where the displacement is being accommodated on thrust faults.

Thirdly, from very long baseline interferometry, Lundgren et al. (1995) determined that the northern Cordillera is moving northeastwards at ~10 mm/a and suggested that most of this motion is being accomodated in the Mackenzie Mountains. From global positioning system (GPS) data, Fletcher and Freymueller (1999) found that Whitehorse is moving at N79°E towards Yellowknife in the North American craton at 5.2 ± 2.3 mm/a (Fig. 1). Using GPS data also, Mazzotti and Hyndman (2002) obtained a vector of 4.5 ± 1.0 mm/a at N20° ± 10°E for the motion of Whitehorse relative to Yellowknife. They speculated that the motion results from the Yakutat block, driven by the Pacific plate, colliding in the Gulf of Alaska with previously accreted terranes of the northern Cordillera. The collision and subduction on north-dipping thrust faults of the Yakutat indenter has caused ~1500 m of additional uplift in the Chugach – St. Elias mountains to the north and dextral transcurrent motion on the Fairweather fault to the east. Following Lewis et al. (2002), Mazzotti and Hyndman (2002) inferred that stress is being transmitted in the upper crust across to the Foreland Belt. They used earthquake data and heat flow data to constrain a kinematic and mechanical model for the region that yields a shortening rate of 4.5 ± 2.5 mm/a across the Mackenzie Mountains — close to the Fletcher and Freymueller (1999) rate of 5.2 ± 2.3 mm/a — and a right-lateral strike-slip rate of 5 ± 2.5 mm/a in the Richardson Mountains to the north. The Beaver River paleopole, by giving a concordant mid-Eocene position, is consistent with the hypothesis that the strain, or decreasing distance between Whitehorse and Yellowknife, is being accommodated to the west of the study area by thrust faulting in the Mackenzie Mountains of the Foreland Belt.

In the northern Foothills of the Foreland Belt, shortening is estimated to be 20–30 km through the upper Paleozoic to Triassic structural levels (Thompson 1979; Gabrielse and Taylor 1982) and ~13 km in Cretaceous strata to the southeast (McMechan 1985). The study area is astride the Triassic–Cretaceous contact, implying that shortening to the east towards Yellowknife is likely between these values, or ~16 ± 4 km. Given such minimal horizontal transport, significant rotation in the study area is highly unlikely, further supporting the notion that the shortening is being accommodated to the west.

Mazzotti and Hyndman (2002) ascribed the eastward motion of the Intermontane terranes to the Yakutat block acting as an indenter. However, between the Yakutat block and the Denali Fault – Coast Shear Zone at the eastern limit of the Intermontane terranes at 60°N are the accreted Chugach, Alexander, and Wrangellia terranes. Carried atop the Kula plate from the south-southwest relative to a fixed North America, these terranes collided in Late Paleocene – Early Eocene with North America along the Denali – Coast Shear Zone (Hillhouse and Coe 1994; Plafker et al. 1994). At ~50 Ma, plate motion in the Gulf of Alaska changed with the onset of Pacific plate subduction. Thus, from ~50 to ~30 Ma, these three terranes would have been driven from the south-southeast against the Intermontane terranes. The arrival of the Yakutat

© 2003 NRC Canada

block, riding on the Pacific plate and colliding against the Chugach terrane along the Fairweather Fault zone, would not have changed significantly the dynamics for the Intermontane terranes. Further, Engebretson et al. (1985, their table 5) showed from sea-floor magnetic anomaly analysis that the Pacific plate has been converging with the North American plate in the Gulf of Alaska since mid-Eocene at about the same angle and velocity as today. This implies that the accreted Intermontane terranes have been obducted eastward onto the North American margin at ~4.8 ± 2.4 mm/a (the average GPS rate) since ~50 Ma, or 240 ± 120 km in total, with the strain accumulated in the Foreland Belt. Consistent with this possibility, the Mackenzie Mountains are being subjected to far more compressional stress than elsewhere in the Foreland Belt. For example, magnitude ≥3.0 earthquakes occur in the Mackenzie Mountains at about five times the rate observed for the Rocky Mountains, despite the chain being only half as long (Sweeney et al. 1992).

Geology and the conundrum

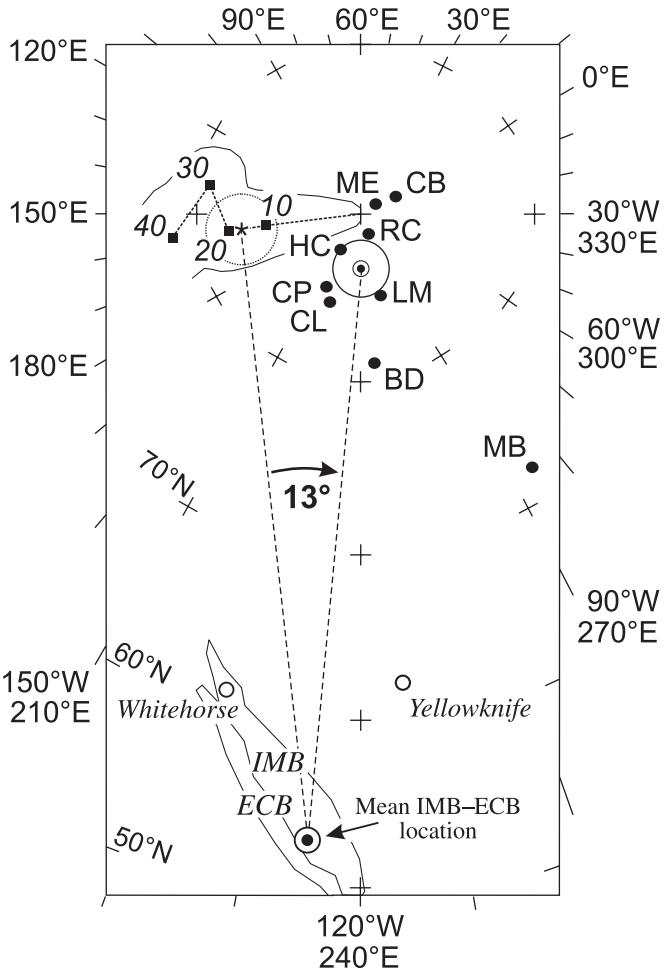
The conundrum arises because there are no known faults west of the Beaver River area that have displacements of the required magnitude to account for convergence of the Intermontane Belt since the Eocene by the 240 ± 120 km that appears to be required by the paleomagnetic data. Using balanced cross-sections, Gordey (1981) estimated that shortening across the entire Foreland Belt in the Mackenzie Mountains was 55 km or less; the Beaver River area is located east of the deformation front, which would reduce even that figure. If the eastward shortening occurred since 42 Ma, Cenozoic right-lateral strike-slip at 5 mm/a in the Richardson Mountains would have amounted to 210 km. Offset of more than a few kilometres on structures in the Richardson Mountains is not allowed by the mapped continuity of strata. Other geological relations in the northern Cordillera similarly seem to preclude large-scale motions on known faults. Thus, there appear to be no obvious structures on which to accommodate the implied rotation as craton-directed contraction even if a rotation axis could be determined.

This inconsistency is reminiscent of the "Baja BC" problem (Irving et al. 1996; Monger and Price 1996; Housen and Beck 1999; Mahoney et al. 1999; Butler et al. 2001), in which paleomagnetic data imply several hundreds to thousands of kilometres of northward latitudinal shift of hinterland parts of the Cordillera in the Cenozoic but geological constraints seem to preclude margin-parallel offsets of such magnitude.

Towards a reconciliation

One possibility to resolve the conundrum outline above is that the relative motions of the Pacific and North American plates were not converging at a consistent, present-day rate since the mid-Eocene, noting that there is debate about the hotspot reference frame on which the relative plate motions are based (eg. Norton 1995; Tarduno and Cottrell 1997). A second possibility is that the convergence component cannot, in fact, be transmitted across the Cordillera to the Mackenzie Mountains. Alternatively, geologic mapping in the Mackenzie Mountains may have missed major thrust faults, noting that such estimates on flat-lying faults are difficult and unreliable, thereby greatly underestimating the amount of shortening. Yet another possibility is that the mid-Eocene paleomagnetic

Fig. 8. Equal-area polar projection showing the 0–40 Ma APWP of Besse and Courtillot (1991) with its 1σ error limit and the Intermontane paleopoles listed in Table 3.



rotation data are a statistical fluke. None of these options are particularly appealing.

If clockwise rotation of the upper crust has occurred in the accreted terranes since 42 Ma, then this bias should appear in younger paleopoles, and it does (Figs. 1, 8; Table 3). One paleopole from the Mount Barr pluton is clearly anomalous (pole MB, Table 3, Fig. 8), but it is based on only five sites and so is excluded from further consideration. Although most were generated some three decades ago, the remaining eight poles are from young unmetamorphosed units that typically are reliable ChRM carriers and unlikely to carry remagnetization components. All eight poles show varying amounts of clockwise rotation relative to their reference pole on the APWP. The mean age of the eight poles is 16 ± 3 Ma and their mean paleopole is located at 87.0°N, 239.4°E (A_{95} = 2.5°), indicating non-significant southward translation of 280 ± 320 km (2.5° ± 2.9°), and significant (2σ) clockwise rotation of 13° ± 5°. Noting that the ages of the paleopoles are fairly uniformly distributed in time from 0 to 42 Ma, two conclusions are evident. First, the upper crust of the Intermontane and eastern Coast Belt terranes has undergone significant clockwise rotation relative to the North American craton; and second, this rotation has been occurring at a rate of ~0.8° ± 0.3°/Ma on average over the past 42 Ma. Two additional features support

© 2003 NRC Canada

Table 3. Post-Middle Eocene paleopoles from the IMB and ECB terranes of the Canadian Cordillera.

	Pole	Age (Ma)	Sites	°N	°E	A_{95}°	Translation (+°N)	Rotation (+°CW)	Reference
ME	Mt. Edziza lavas	0–6	84	89.1°	31.8	6.6°	1° ± 6°	5° ± 10°	Souther and Symons 1974
LM	Level Mountain lavas	1–7	38	85.0°	256.0	7.2°	–4° ± 6°	11° ± 13°	Hamilton and Evans 1983
CP	Cariboo mafic plugs	5–10	17	84.9°	213.3	5.2°	–5° ± 5°	4° ± 8°	Symons 1969*a*
CL	Cariboo lavas	8–15	48	84.0°	219.3	5.1°	–6° ± 5°	7° ± 8°	Symons 1969*c*
BD	Brown mafic dikes	10–15	9	81.7°	247.8	16.2°	–7° ± 13°	16° ± 28°	Symons 1969*b*
RC	Rogers Creek pluton	15–17	11	88.3°	257.5	5.7°	–1° ± 5°	13° ± 8°	Vandall and Wheadon 1992
MB	Mount Barr pluton*	16–21	5	72.4°	274.2	14.0°			Symons 1973
CB	Chiliwack batholith	30–37	15	87.5°	9.5	2.2°	–1° ± 3°	17° ± 5°	Beck et al. 1982
HC	Hope complex	34–42	5	87.7°	208.1	5.6°	–2° ± 5°	14° ± 8°	Symons 1973
	Mean IMB-ECB	**16 ± 3**	–8	87.0°	239.4	2.5°			

Note: Pole abbreviations are defined in Fig. 1. Sites indicates the number of sampling sites contributing to the reported paleomagnetic result. Pole position is given as °N, north latitude (in degrees); °E, east longitude (in degrees); A_{95}°, radius of cone of 95% confidence (in degrees). * denotes exclusion of the anomalous pole from calculation of the Intermontane Belt - eastern Coast Belt (IMB-ECB) mean paleomagnetic pole.

these conclusions. First, the movement direction is nearly perpendicular to the N30° ± 10°W trending long axis of these terranes during the Neogene (Fig. 8) and is similar to the determined GPS compression directions of N79°E and N20° ± 10°E (Fletcher and Freymuellller 1999; Mazzotti and Hyndman 2002). Second, the determined rotation since 16 Ma of 13° ± 5° is proportional within errors to the rotation since 51 Ma of 24° ± 10° (Symons et al. 2000), indicating a systematic tectonic response since mid-Eocene of these upper crustal terranes from Washington to the Yukon. This implies, in turn, that the Intermontane and eastern Coast Belt terranes have responded to the tectonic stresses as a quasi-rigid, quasi-coherent sheet as their morphogeological features imply.

It is important to recognize that the utilized paleomagnetic data base includes data from 15 collections spread throughout the Intermontane terranes that incorporate > 400 sites from rock units with reliable ChRM determinations and known attitudes that were measured in several different laboratories over nearly four decades, and every collection shows clockwise rotation regardless of its age. The geophysical evidence for post mid-Eocene motion in the Yukon is substantial and must be accounted for.

Two recent interpretations indicate the extent of the North American craton under the accreted terranes of the northern Cordillera. Snyder et al. (2002) interpreted the SNORCLE 2 and 3 deep crustal seismic profiles to show that Proterozoic cratonic rocks extend southwesterly from outcrops in the Mackenzie Mountains, across the Omineca Belt under a thin skin of Paleozoic strata, and then ramp down under the accreted terranes of the Intermontane Belt to pinch out under the eastern Coast Belt close to the Denali – Coast Shear Zone fault. In this interpretation, "…some terranes such as the Yukon–Tanana, Cache Creek, and Quesnellia were detached from their roots and thrust 200 to 400 km onto the margin of North America as thin (<5 km) crustal flakes…" whereas "…other terranes, such as Stikinia, docked above and outside of the leading edge of the layered wedge and make up most of the crust today". They used an estimate of 450 to 800 km Early Cretaceous – Oligocene dextral strike slip on the Tintina and related faults, but only ≤125 km of this is Late Eocene to Oligocene motion (Gabrielse 1985). The preservation of the Tintina fault zone as a crustal-scale vertical structure cutting across the overriding "thin crustal flakes" implies that the fault zone cuts into the underlying Proterozoic rocks and that vertical motions and related erosion are important for survival of its topographic expression. Note, however, that when the geometry and current 5 ± 2.5 mm/a rate of northward motion in the Richardson Mountains are considered over the past 42 Ma only, a dextral motion of ~210 ± 105 km is implied for the Tintina fault zone.

In contrast Abraham et al. (2001) and Francis (2002) interpreted geochemical and isotopic characteristics of Cretaceous to Recent volcanic centres in the northern Cordillera to place the cratonic margin close to the Omineca and Intermontane Belt boundary (Fig. 1). They also interpreted data from mid-Tertiary and younger centres such as Mt. Edziza and Level Mountain to show that the transition from lithospheric mantle of the Coast Belt to Intermontane Belt composition is now displaced ~150 km eastwards from their surface contact. Alternatively, the eastward motion of a thin crustal flake would have severed the volcanic centres from their mantle roots, in which case the ~150 km displacement is simply another measure of eastward motion of the flake over the Proterozoic ramp since the mid-Tertiary. Similarly, the systematic geochemical and isotopic changes observed in the ~70 Ma Carmacks Group volcanics near the boundary between the Omineca and Intermontane Belts could mark ~200 km of eastwards translation over the Proterozoic ramp from its western edge since ~70 Ma.

We speculate that the Intermontane terranes are rotating on top of a basement that includes the Yukon–Tanana terrane and underlying Proterozoic cratonic wedge. If true then only some of the eastward compression recorded by the paleomagnetic data needs to be accommodated by thrust faults in the Mackenzie Mountains, i.e., some motion could be accommodated by new or reactivated thrust and sole faults in the Selwyn Mountains and Yukon–Tanana terrane, akin to the collapsing of a deck of cards. Another consideration is that the pivot point for rotation of the Intermontane need not be in southern British Columbia; it could be in the middle of north-central British Columbia. In this case Eocene and younger extension in the Omineca Belt of southern British

© 2003 NRC Canada

Columbia will also result in clockwise rotation. Here, Parrish et al. (1992) have reported 90 ± 10 km of extension coinciding with intrusion of the 52 ± 6 Ma Coryell intrusions, and Souther and Yorath (1992) have documented substantial Miocene and Pliocene volcanic effusions with related extension in the southern Intermontane. A further option is that some clockwise rotation may be accommodated by segmentation of the Intermontane terranes as a result of greater dextral motion on the Denali – Coast Shear Zone than on the Tintina – Rocky Mountain Trench Fault, but the evidence for the necessary intervening east–west faults to accommodate such segmentation is poor.

Conclusions

The paleomagnetic pole obtained for the Beaver River Alkalic Complex is of high quality with $Q = 7$. It is dated precisely, and it is supported by several times the minimum number of specimens with much better than minimal cluster estimates, thorough AF and thermal step demagnetization with alignment of subtracted ChRM vectors, and a positive contact test on the Larson Creek West pluton, thus satisfying Q criteria 1 to 4. Combined stratigraphic and radiometric evidence shows that the intrusions postdate regional tectonic deformation to satisfy criterion 5. Also the three plutons give the same ChRM direction, excluding polarity, despite being emplaced into host rocks with significantly differing ages and attitudes, and the host rock at site 27 defines an impossible virtual pole position for the Eocene on tilt correction. The presence of reversals both within the Larson Creek West pluton and between the plutons satisfies criterion 6. Finally, criterion 7, designed to guard against post-emplacement remagnetization, requires that the paleopole does not resemble a younger age by more than one geological period, in this case a Quaternary paleopole of less than 1.6 Ma. A fair comparison would be the 0 to 5 Ma pole of McElhinny et al. (1996). Their pole for 4408 sites in 0 to 5 Ma igneous rocks with normal and reversed polarities is located at 89°N, 143°E (A_{95} = 0.8°). Noting that its cone of 95% confidence does not intersect the Beaver River cone (Fig. 7), and that the well-known far-sided and right-handed effect (McElhinny and Merrill 1975) would increase separation between these two poles, criterion 7 is easily met by the Beaver River paleopole. Thus, we conclude that this paleopole provides a highly reliable North American contribution to APWP constructions.

The conformable paleopole of the 42 Ma Beaver River alkalic complex contrasts with the paleopoles from six ~50 Ma Eocene units in the accreted Intermontane and eastern Coast Belt terranes. The latter poles on average record no northward translation but 24° ± 10° of clockwise rotation (Symons et al. 2000). In particular, the White Pass dikes, due west of the study area at ~60°N, record 40° ± 9° of clockwise rotation. Also eight poles from these same terranes with an average age of 16 ± 3 Ma record non-significant northward translation, but 13° ± 5° of clockwise rotation that is equivalent to eastward translation. This contrast with the concordant Beaver River pole likely results from the upper crust, composed of the accreted terranes and pushed by the subducting Pacific plate, sliding 240 ± 120 km eastward as a thin skin on a décollement over a weaker lower crust, composed of the pericratonic Yukon–Tanana terrane and Proterozoic wedge, on the western margin of the North American craton. As the upper crust slid onto and up the margin it met increasing resistance, and it is inferred from the geometry that the accreted terranes rotated preferentially clockwise, that the strain was accommodated substantially on thrust faults in the Mackenzie Mountains just west of the Beaver River study area, as well as in the underlying crust. The puzzle from a geological perspective is to identify faults with sufficient displacements or other mechanisms to account for the magnitude of the proposed post mid-Eocene motion.

Acknowledgments

The authors wish to thank pilots Dan Beaulac and Lance White of Deh Cho Helicopters, Fort Liard, N.W.T., for safe and effective helicopter support; Jaros Kulhanek who made the remanence measurements; and Michael Harris who ran the component analysis and saturation remanence calculations. We are grateful to Don Francis of McGill University, Montréal, and S.M. Rowins of The University of British Columbia, Vancouver, B.C., who provided logistical and geological information as well as unpublished Ar^{40}/Ar^{39} dates. This manuscript has greatly benefitted from the comments of reviewers Grant Abbott of the Yukon Geology Program and David Stone of the University of Fairbanks, Fairbanks, as well as those of the Associate Editor, Fred Cook. This work was supported by Natural Sciences and Engineering Research Council research grants to PE (705-00) and DTAS (7834-00).

References

Abraham, A.C., Francis, D., and Polvé, M. 2001. Recent alkaline basalts as probes of the lithospheric mantle roots of the Northern Canadian Cordillera. Chemical Geology, **175**: 361–386.

As, J.A., and Zijderveld, J.D.A. 1958. Magnetic cleaning of rocks in paleomagnetic research. Geophysical Journal of the Royal Astronomical Society, **1**: 308–319.

Bailey, R.C., and Halls, H.C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

Bardoux, M., and Irving, E. 1989. Paleomagnetism of Eocene rocks of the Kelowna and Castlegar areas, British Columbia: studies in determining paleohorizontal. Canadian Journal of Earth Sciences, **26**: 829–844.

Beck, M.E., Burmester, R.F., and Schoonover, R. 1982. Tertiary paleomagnetism of the north Cascade Range, Washington. Geophysical Research Letters, **9**: 515–518.

Besse, J., and Courtillot, V. 1991. Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma. Journal of Geophysical Research, **96**: 4029–4050.

Butler, R.F., Gehrels, G.E., and Kodama, K.P. 2001. A moderate translation alternative to the Baja British Columbia hypothesis. GSA Today, (6): 4–10.

Cande, S.C., and Kent, D.V. 1995. Revised calibration of the geomagnetic polarity timescale for the Late Cretaceous and Cenozoic. Journal of Geophysical Research, **100**: 6093–6095.

Cook, F.A. 1995. Lithosperic processes and production in the southern Canadian Cordillera: a Lithoprobe perspective. Canadian Journal of Earth Sciences, **32**: 1803–1824.

Currie, K.L., and Larochelle, A. 1969. A paleomagnetic study of

© 2003 NRC Canada

volcanic rocks from Mistastin Lake, Labrador, Canada. Earth and Planetary Science Letters, **6**: 309–315.

Douglas, R.J.W. 1976. La Biche River, District of Mackenzie. Geological Survey of Canada, Map 1308A, scale 1 : 250 000.

Engebretson, D.C., Cox, A., and Gordon, R.G. 1985. Relative plate motions between oceanic and continental plates in the Pacific Basin. Geological Society of America, Special Paper 206.

Everitt, C.W.F., and Clegg, J.A. 1962. A field test of paleomagnetic stability. Geophysical Journal of the Royal Astronomical Society, **6**: 312–319.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, **A217**: 295–305.

Fletcher, H.J., and Freymueller, J.T. 1999. New GPS constraints on the motion of Yakutat block. Geophysical Research Letters, **26**: 3029–3032.

Flynn, J.J. 1986. Correlation and geochronology of Middle Eocene strata from the western United States. Palaeogeography, Palaeoclimatology, Palaeoecology, **55**: 335–406.

Fox, K.F., and Beck, M.E., Jr. 1985. Paleomagnetic results for Eocene volcanic rocks from northeastern Washington and the Tertiary tectonics of the Pacific Northwest. Tectonics, **4**: 323–341.

Francis, D. 2002. Interrogating the lithosphere of the Northern Canadian Cordillera with Cretaceous to Recent volcanics and mantle xenoliths. Lithoprobe Report, **82**: 24–30.

Gabrielse, H. 1985. Major dextral transcurrent displacements along the northern Rocky Mountain Trench and related lineaments in north-central British Columbia. Geological Society American Bulletin, **96**: 1–14.

Gabrielse, H., and Taylor, G.C. 1982. Geological maps and cross-sections of the Cordillera from near Fort Nelson, British Columbia to Gravina Island, southeastern Alaska. Geological Survey of Canada, Open File 864.

Gordey, S.P. 1981, Structure sections across south-central Mackenzie Mountains, N.W.T. Geological Survey of Canada, Open File 809.

Hamilton, T.S., and Evans, M.E. 1983. A magnetostratigraphic and secular variation study of Level Mountain, northern British Columbia. Geophysical Journal of the Royal Astronomical Society, **73**: 39–49.

Harris, M.J., Symons, D.T.A., Blackburn, W.H., and Hart, C.J.R. 1999. Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne stock, southern Yukon Territory. Canadian Journal of Earth Sciences, **36**: 905–915.

Hillhouse, J.W., and Coe, R.S. 1994. Paleomagnetic data from Alaska. *In* The Geology of Alaska. *Edited by* G. Plafker and H.C. Berg. The Geology of North America, Decade of North American Geology, Geological Society of America, Boulder, Colo., Vol. G-1, pp. 797–812.

Housen, B.A., and Beck, M.E. 1999. Testing terrane transport: An inclusive approach to the Baja B.C. controversy. Geology, **27**: 1143–1146.

Irving, E., Wynne, P.J., Thorkelson, D.J., and Schiarizza, P. 1996. Large (1000 to 4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. Journal of Geophysical Research, **101**: 17 901–17 916.

Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J.R., Enkin, R.J., and Engebretson, D.C. 1996. Yellowstone in Yukon: the Late Cretaceous Carmacks Group. Geology, **24**: 997–1000.

Kirschvink, J.L. 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Lewis, T.J., and Hyndman, R.D. 2001. High heat flows and high crustal temperatures along the SNORCLE transect. Lithoprobe Report 79, pp. 28–29.

Lewis, T.J., Hyndman, R.D., and Flueck, P. 2002. Thermal controls on present tectonics in the northern Canadian Cordillera. Lithoprobe Report 82, pp. 15–16.

Lovlie, R., and Opdyke, N.D. 1974. Rock magnetism and paleomagnetism of some intrusions from Virginia. Journal of Geophysical Research, **79**: 343–349.

Lundgren, P., Saucier, F., Palmer, R., and Langon, M. 1995. Alaska crustal deformation: Finite element modelling constrained by geologic and very long baseline interferometry data. Journal of Geophysical Research, **100**: 22 033 – 22 045.

Mahoney, J.B., Mustard, P.S., Haggart, J.W., Friedman, R.M., Fanning, C.M., and McNicoll, V.J. 1999. Archean zircons in Cretaceous strata of the western Canadian Cordillera: The "Baja B.C". hypothesis fails a "crucial test". Geology, **27**: 195–198.

Mazzotti, S., and Hyndman, R.D. 2002. Yakutat collision and strain transfer across the northern Canadian Cordillera. Geology, **30**: 495–498.

McElhinny, M.W., and McFadden, P.L. 2000. Paleomagnetism: Continents and Oceans. International Geophysics Series, 73. Academic Press, San Diego, Calif.

McElhinny, M.W., and Merrill, R.T. 1975. Geomagnetic secular variation over the last 5 m.y. Reviews in Geophysics, **13**: 687–708.

McElhinny, M.W., McFadden, P.L., and Merrill, R.T. 1996. The time-averaged paleomagnetic field 0–5 Ma. Journal of Geophysical Research, **101**: 25 007 – 25 027.

McMechan, M.E. 1985. Low-taper triangle zone geometry: an interpretation for the Rocky Mountain Foothills, Pine Pass – Peace River area, British Columbia. Bulletin of Canadian Petroleum Geology, **33**: 31–38.

McMechan, M.E., and Thompson, R.I. 1989. Structural style and history of the Rocky Mountain Fold and Thrust Belt *In* Western Canada Sedimentary Basin: A case history, *Edited by* B.D. Ricketts. Canadian Society of Petroleum Geologists, pp. 41–47.

Monger, J.W.H., and Price, R.A. 1996. Comment on "Paleomagnetism of the Upper Cretaceous strata of the Mt. Tatlow: Evidence for 3000 km of northward displacement of the eastern Coast Belt" by P.J. Wynne et al., and on "Paleomagnetism of the Spences Bridge Group and northward displacement of the Intermontane Belt, British Columbia: A second look" by E. Irving et al. Journal of Geophysical Research, **101**: 13 793–13 799.

Norton, I.O. 1995. Plate motions in the North Pacific: The 43 Ma nonevent. Tectonics, **14**: 1080–1094.

Parrish, R.R., Friedman, R.M., and Armstrong, R.L. 1992. Eocene extension faults. *In* Geology of the Cordilleran Orogeny in Canada. *Edited by* H. Gabrielse and C.J. Yorath. Geological Survey of Canada, no. 4, pp. 660–674.

Pigage, L.C., and Allen, T.L. 2001. Geological map of Pool Creek (NTS 95C/5), southeastern Yukon (1 : 50,000 scale), Exploration and Geological Services Division, Yukon Region, Indian and Northern Affairs Canada, Open File 2001-32.

Plafker, G., Moore, J.C., and Winkler, G.R. 1994. Geology of the southern Alaska margin, *In* The Geology of Alaska. *Edited by* G. Plafker and H.C. Berg. The Geology of North America, Decade of North American Geology, Geological Society of America, Boulder, Colo., Vol. G-1, pp. 389–449.

Ressetar, R., and Martin, D.L. 1980. Paleomagnetism of Eocene igneous intrusives in the Valley and Ridge Province, Virginia and West Virginia. Canadian Journal of Earth Sciences, **17**: 1583–1587.

Rowins, S.M., and Francis, D. 1995. The Late Tertiary to Recent Beaver River alkaline complex, southeastern Yukon, Canada. Geological Society of America, Abstracts with Programs: A-46.

Sheriff, S.D., and Shive, P.N. 1980. The Rattlesnake Hills of central Wyoming revisited: further paleomagnetic results. Geophysical Research Letters, **7**: 589–592.

© 2003 NRC Canada

Shive, P.N., and Pruss, E.F. 1977. A paleomagnetic study of basalt flows from the Absaroka Mountains, Wyoming. Journal of Geophysical Research, **82**: 3039–3048.

Snyder, D.B., Clowes, R.M., Cook, F.A., Erdmer, P., Evenchick, C.A., van der Velden, A.J., and Hall, K.W. 2002. Proterozoic prism arrests suspect terranes: Insights into the Ancient Cordilleran margin from seismic reflection data. GSA Today, **12**(10): 4–10.

Souther, J.G., and Symons, D.T.A. 1974. Stratigraphy and paleomagnetism of the Mount Edziza volcanic complex, British Columbia. Geological Survey of Canada, Paper 73-72.

Souther, J.G., and Yorath, C.J. 1992. Neogene assemblages. *In* Geology of the Cordilleran Orogeny in Canada. *Edited by* H. Gabrielse and C.J. Yorath. Geological Survey of Canada, no. 4, pp. 375–401.

Sweeney, J.F., Stephenson, R.A., Currie, R.G., and DeLaurier, J.M. 1992. Crustal Geophysics. *In* Geology of the Cordilleran Orogeny of Canada. *Edited by* H. Gabrielse and C.D. Yorath. Geological Survey of Canada., no. 4: pp. 39–59.

Symons, D.T.A. 1969*a*. Paleomagnetism of four Late Miocene gabbroic plugs in south-central British Columbia. Canadian Journal of Earth Sciences, **6**: 653–662.

Symons, D.T.A. 1969*b*. Geological implications of paleomagnetic studies in the Bella Coola and Laredo Sound map areas, British Columbia (93D and 103A). Geological Survey of Canada, Paper 68-72.

Symons, D.T.A. 1969*c*. Paleomagnetism of the Late Miocene plateau basalts in the Cariboo region of British Columbia. Geological Survey of Canada, Paper 69-43.

Symons, D.T.A. 1973. Paleomagnetic results from the Tertiary Mount Barr and Hope plutonic complexes, British Columbia. Geological Survey of Canada Paper 73-19.

Symons, D.T.A. 1974. Age and tectonic implications of paleomagnetic results from plutons near Prince Rupert, British Columbia. Journal of Geophysical Research, **79**: 2690–2697.

Symons, D.T.A. 1977. Geotectonics of Cretaceous and Eocene plutons in British Columbia, a paleomagnetic fold test. Canadian Journal of Earth Sciences, **26**: 821–828.

Symons, D.T.A., and Wellings, M.R. 1989. Paleomagnetism of the Eocene Kamloops Group and the cratonization of Terrane I of the Canadian Cordillera. Canadian Journal of Earth Sciences, **26**: 821–828.

Symons, D.T.A., Harris, M.J., Gabites, J.E., and Hart, C.J.R. 2000. Eocene (51 Ma) end to northward translation of the Coast Plutonic Complex: paleomagnetism and K–Ar dating of the White Pass dikes. Tectonophysics, **326**: 93–109.

Tarduno, J.A., and Cottrell, R.D. 1997. Paleomagnetic evidence for motion of the Hawaiian hotspot during the formation of the Emperor seamounts. Earth and Planetary Science Letters, **153**: 171–180.

Thompson, R.I. 1979. A structural interpretation across part of the northern Rocky Mountains, British Columbia. Canadian Journal of Earth Sciences, **16**: 1228–1241.

Torsvik, T.H., Mosar, J., and Eide, E.A. 2001*a*. Cretaceous–Tertiary geodynamics: A North Atlantic exercise. Geophysical Journal International, **146**: 850–866.

Torsvik, T.H., Van der Voo, R., Meert, J.G., Mosar, J., and Walderhaug, H.J. 2001*b*. Reconstructions of the continents around the North Atlantic at about the 60th parallel. Earth and Planetary Science Letters, **187**: 55–69.

Van der Voo, R. 1993. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans. Cambridge University Press, Cambridge, U.K.

Vandall, T.A., and Palmer, M.C. 1990. Upper limit of docking time for Stikinia and Terrane I: paleomagnetic evidence from the Eocene Ootsa Lake Group, British Columbia. Canadian Journal of Earth Sciences, **27**: 212–218.

Vandall, T.A., and Wheadon, P.M. 1992. Are post-Middle Miocene tilts present in the southern Coast Belt of British Columbia? A case study of present Earth's magnetic field overprints. Canadian Journal of Earth Sciences, **29**: 2370–2382.

© 2003 NRC Canada